TECHNICAL BULLETIN | ROBOTICS · 4WD CHASSIS | REFERENCE ARCHITECTURE

Bluetooth RC Car — Wiring & Connection Guide

Hardware Interface Specification, Software Serial Protocols, and Power Isolation

CRUCIAL INTEGRATION NOTICE: L293D shields are produced in diverse hardware topologies. This document establishes an engineering baseline. Never assume pin parity across clones without verifying traces.

1. System Summary & Objectives

This engineering document provides a robust electrical connection blueprint for a 4-wheel-drive robotic car utilizing an Arduino Uno R3, an L293D motor driver shield, and an HC-05/06 Bluetooth transceiver. The architecture enforces complete electrical decoupling between micro-controller logic and high-inductive motor EMF, preventing resets during in-rush motor current transients.

Key System Ratings

4.5–9V	600 mA	3.3V
MOTOR RAIL (VM)	CONTINUOUS/CH	BT RX PEAK

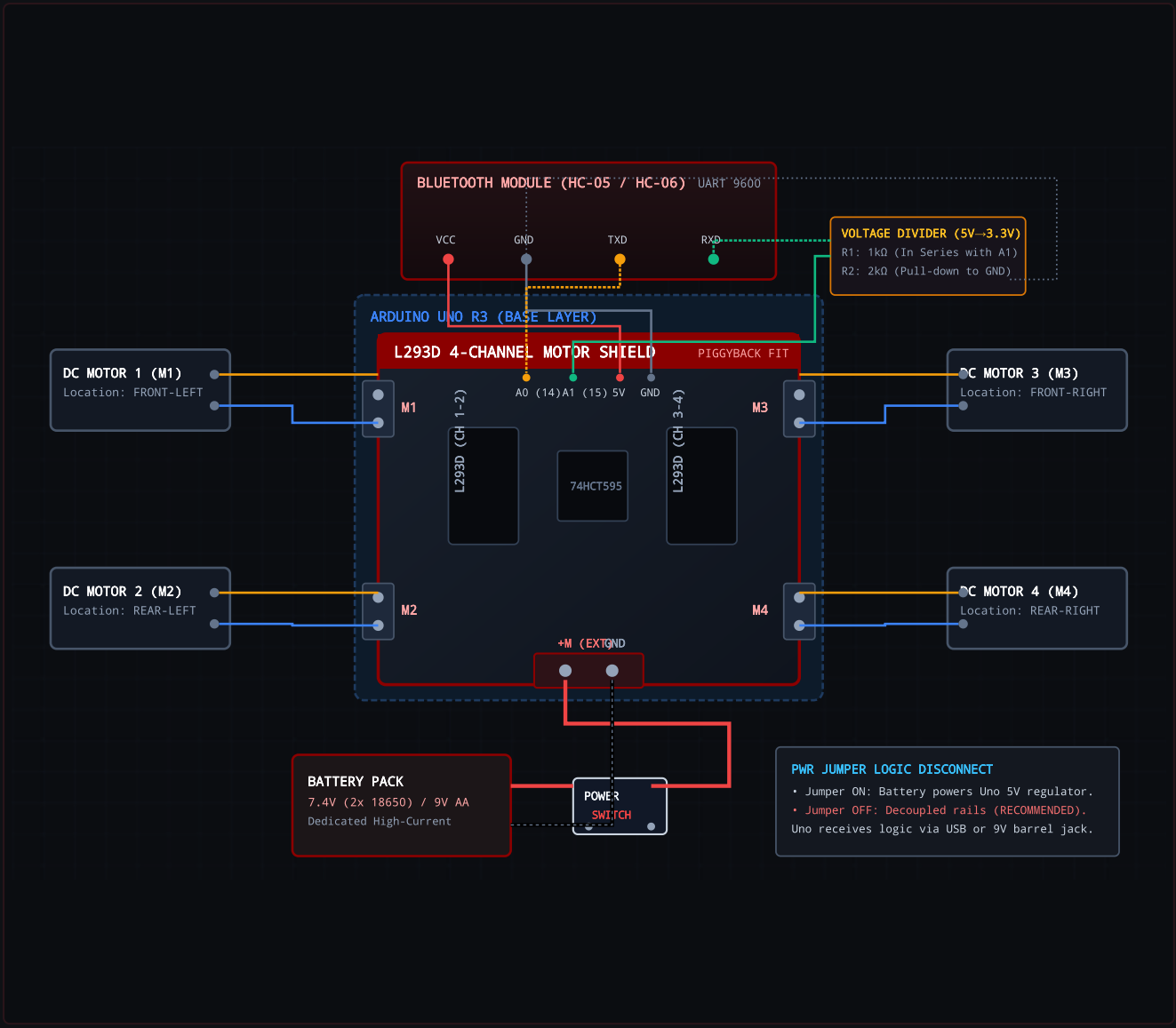
2. BILL OF MATERIALS & HARDWARE VERIFICATION

#	COMPONENT / MODULE	ENGINEERING SPEC / MODEL	CRITICAL FUNCTION & VERIFICATION NOTES
01	Microcontroller Core	Arduino Uno R3 (ATmega328P)	5V TTL logic level. Provides PWM timing and executes command dispatch.
02	Quad Motor Driver Shield	L293D Dual H-Bridge Shield	Plugs onto Uno headers. Drives 4 DC motors via internal 74HCT595 shift register.
03	Drive Actuators (4x)	TT Gearbox DC Motors (1:48)	Rated 3–6V DC. Install 0.1µF ceramic noise decoupling capacitors across terminals.
04	Platform & Wheels	4WD Robotic Chassis Base	Independent wheel drives: Left Bank (M1/M2) and Right Bank (M3/M4).
05	Wireless Transceiver	HC-05 (Master/Slave) or HC-06	Default baud 9600. Backplane regulator requires 3.6–6V VCC, but RX requires 3.3V logic.
06	Primary Motor Battery	2x 18650 Li-Ion (7.4V) or 6x AA	Supplies high surge current directly to motor shield EXT_PWR. Never draw from Uno 5V!
07	Secondary Logic Battery	9V Alkaline or USB 5V (Optional)	Permits split-rail power topology when the motor shield PWR jumper is removed.
08	Master Isolation Switch	SPST / Rocker Toggle (3A rated)	Installed directly inline with positive battery rail to safely isolate raw energy source.
09	Signal Level Attenuator	1kΩ & 2kΩ Resistors (1/4W)	Forms a passive voltage divider bridging Arduino 5V TX to HC-05 3.3V RX terminal.
10	Wiring Interconnects	22–24 AWG Solid / Stranded Wire	Color-coded high-flex jumper cables for vibration resistance and field debugging.

3. COMPLETE INTERCONNECT & WIRING DIAGRAM

REFERENCE CONFIGURATION · VERIFY SHIELD PINOUT

The schematic below illustrates the complete four-motor connection topology, battery isolation circuit, and level-shifted wireless communications link. Motors M1 and M2 govern the left track; M3 and M4 govern the right track.



WIRING STANDARD: SOLID RED Batt + / 5V Rail · BLACK Batt / Chassis Ground · YELLOW Serial TXD (A0 / D14) · GREEN Serial RXD (A1 / D15) · BLUE Motor Terminal Pair

4. PIN-TO-PIN INTERFACE & CHANNEL ALLOCATION

Software Serial Mapping Note: On the ATmega328P (Uno R3), analog pins A0 and A1 execute digital I/O operations as digital pins 14 and 15 respectively. The instantiation `SoftwareSerial bluetoothSerial(14, 15);` defines **Pin 14 (A0) as Microcontroller RX** and **Pin 15 (A1) as Microcontroller TX**.

Table 4.1: DC Motor Driver Channel Allocations

ACTUATOR ID	CHASSIS POSITION	SHIELD TERMINAL	CONTROL REGISTER / TIMER	OPERATIONAL & POLARITY NOTES
Motor 1	Front Left (FL)	`M1 Screws`	74HCT595 (Q2, Q3) · PWM D11	If forward code drives wheel backwards, invert wire pair polarity at M1 terminal.
Motor 2	Rear Left (RL)	`M2 Screws`	74HCT595 (Q1, Q4) · PWM D3	Paired in software logic with M1 for differential tank steering.
Motor 3	Front Right (FR)	`M3 Screws`	74HCT595 (Q5, Q7) · PWM D6	Right bank requires reversed rotational polarity relative to left bank.
Motor 4	Rear Right (RR)	`M4 Screws`	74HCT595 (Q0, Q6) · PWM D5	Paired in software logic with M3. Verify concurrent torque output.

Table 4.2: HC-05 / HC-06 Bluetooth Transceiver Connections

MODULE PIN	TARGET CONNECTION	SIGNAL DOMAIN	ELECTRICAL DIRECTION	ENGINEERING CONSTRAINT & LEVEL SHIFT RATIONALE
VCC	`Arduino 5V`	Power Supply	Power Source (IN)	Feeds module onboard low-dropout 3.3V LDO regulator. Minimum input 3.6V.
GND	`System GND`	Common Ground	Reference (COMMON)	Essential reference loop for both RF baseband and high-frequency digital UART.
TXD	`Pin 14 (A0)`	UART 3.3V TTL	BT (OUT) → Uno (IN)	Direct link safe: Uno recognizes 3.3V high as reliable TTL logic HIGH (>3.0V).
RXD	`Pin 15 (A1)`	UART Level Shift	Uno (OUT) → BT (IN)	**MANDATORY ATTENUATION:** Uno 5V output will degrade 3.3V CSR radio chip. Attenuate via 1kΩ series / 2kΩ pull-down network.
STATE / EN	`N/C (Unused)`	Mode Control	Float / Open	Leave unconnected for standard operational slave streaming.

Table 4.3: Power Isolation & Distribution Bus

POWER RAIL	PHYSICAL TERMINATION	TARGET VOLTAGE	DECOUPLING & JUMPER MANAGEMENT
Motor Rail (VM)	`EXT_PWR (+ / -)`	6.0V – 9.0V DC	Connect raw battery output via master switch. Maximum continuous capacity 1.2A total.
Shield PWR Jumper	`2-Pin Header`	Bridging Link	**REMOVE JUMPER** to isolate logic supply from inductive spikes during motor stalls.
Logic Rail (VCC)	`Uno USB / DC Jack`	5.0V Regulated	Independent 9V battery or power bank USB line maintains clean logic rails during operation.

5. COMMISSIONING, SOFTWARE COMPATIBILITY & VALIDATION

Assembly Execution Sequence

1. Seat L293D shield onto Uno R3, confirming zero bent header pins.
2. Solder 0.1µF ceramic caps across motor tabs to suppress brush RF noise.
3. Fasten motors to chassis; terminate leads into M1–M4 terminals.
4. Assemble 1kΩ / 2kΩ divider circuit; wire HC-05 TXD/RXD to A0/A1.
5. Remove PWR jumper; wire external battery via rocker toggle.

Firmware Architecture & Syntax

Include required hardware driver and communication headers:

```
#include <AFMotor.h>
#include <SoftwareSerial.h>
// Pin 14 = Analog A0 (RX), Pin 15 = Analog A1 (TX)
SoftwareSerial bluetoothSerial(14, 15);
AF_DCMotor motor1(1); AF_DCMotor motor2(2);
AF_DCMotor motor3(3); AF_DCMotor motor4(4);
```

LIBRARY COMPATIBILITY AUDIT (CRITICAL)

The `AFMotor.h` library exclusively commands Adafruit V1-style motor shields utilizing a 74HCT595 serial-to-parallel shift register. If your clone shield utilizes dedicated discrete H-bridge pins (e.g., L298N shields or direct PWM pin bridges), `AFMotor` will fail to spin motors and may hold A0/A1 low. Always verify onboard IC markings prior to flashing.

Pre-Flight Bench Testing Procedure (Wheels Suspended)

Phase 1: Zero-Load Verification. Mount vehicle chassis on a bench stand so all four wheels rotate freely without ground contact. Connect Uno via USB with battery toggle OFF.
Phase 2: RF Link Handshake. Power module. Verify HC-05 LED blinks rapidly (2 Hz). Pair via mobile serial terminal (Passkey: 1234 or 0000). LED transitions to slow pulse.
Phase 3: Actuator Phasing. Switch battery ON. Transmit forward drive token (`'F'`). Verify all four wheels turn forward. If M1 rotates reverse, swap its terminal pair.

Table 5.1: Fault Isolation & Troubleshooting Matrix

SYMPTOM	ROOT CAUSE HYPOTHESIS	CORRECTIVE ENGINEERING ACTION
Uno Resets Under Load	Brownout caused by in-rush motor current drawn from Uno 5V.	Remove shield PWR jumper. Provide discrete battery pack directly to EXT_PWR.
No Bluetooth Comm	TX/RX reversed or baud rate mismatch in `SoftwareSerial`.	Confirm crossover (Uno TX→Module RX; Uno RX→Module TX). Confirm 9600 baud rate.
Motor Whines / Stalls	Supply rail voltage dropped below L293D internal diode threshold.	Recharge Li-Ion cells. Ensure stall current does not exceed 1.2A thermal trip point.

Safety & Operational Safeguard: Never switch on power while adjusting bare terminal blocks. Continuous stall events over 3 seconds can permanently destroy the L293D silicon driver stages. Always implement a dead-man stop routine in software that commands all channels RELEASE if Bluetooth connectivity is dropped.